



**Verified Carbon
Standard**

IMPROVED COOKSTOVE DISTRIBUTION IN GHANA BY EKI



EKI Energy Services Limited

Project Title	Improved Cookstove Distribution in Ghana by EKI
Version	03
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1 PROJECT DETAILS

1.1 Summary Description of the Project

The main purpose of the project is the distribution of fuel-efficient improved cook stoves (ICS) in the Republic of Ghana (Ghana). The Improved cookstove stove distributed through this project has replaced the low-efficient baseline cookstove.

In rural Ghana, a large portion of households rely on charcoal-based mud or three-stone stoves for daily cooking. In addition to having a relatively low thermal efficiency, these mud or three-stone stoves also contribute to health risks from indoor air (smoke) pollution.

Through this project, the distribution, and installation of high thermal efficient improved cook stoves (ICS) has been undertaken for households. The ICS burns charcoal more efficiently thereby improving thermal energy transfer to pots, hence saving charcoal and greenhouse gases. Not only will this halt the rapidly progressing deforestation but also reduce health hazards from indoor air (smoke) pollution and time & money of participating household now have to spend less time in the procurement of charcoal.

As the target populations are unable to afford these stoves (ICS), project promoters have distributed ICS free of cost. The end-user has been informed in advance that the use of ICS generates carbon finance which in turn is used to cover the price of ICS and for recovering project implementation costs.

The project locations for initial Project Activity Instances are in Republic of Ghana (Ghana) and it may expand further in other parts of Africa. EKI has distributed 20,102 ICS till the end of this monitoring period i.e., 28-February-2023.

EKI has considered one cookstove as one project activity instance for this group project activity. The first project activity instance was implemented on 16-January-2023 under this grouped project activity.

The average annual GHG emission reductions for the group project activity are expected to be 72,030 tCO₂e, and hence, total GHG emission reductions over the entire crediting period are expected to be 504,210 tCO₂e.

The 1st monitoring survey for the grouped project activity was conducted from 16-January-2023 to 28-February-2023. The project results in a total emission reduction of 4,291 tCO₂e over the monitoring period of 16-January-2023 to 28-February-2023 (inclusive of both the dates).

1.2 Sectoral Scope and Project Type

The sectoral scope and type of project applicable are as below:

Sectoral scope: 03 - Energy demand

Type: II – Energy efficiency improvement projects

The project is a grouped project

1.3 Project Eligibility

The project involves energy-efficient cookstove distribution which falls under the category of efficiency improvements in the thermal application, therefore it is eligible under the scope of the VCS Program. As per VCS Standard Version 4.4, section 2.1.3, efficiency improvements in thermal applications (e.g., cook stoves) are not excluded, hence the project is eligible under VCS.

1.4 Project Design

The current project activity is a grouped project.

Eligibility Criteria

For the inclusion of new project activity instances i.e., ICS, the project proponent shall ensure that it meets the eligibility criteria below:

No.	Criterion	How the new project activity instances comply
1	Meet the applicability conditions set out in the methodology applied to the project	New project activity instances (Energy Efficient Cook Stoves) will meet the applicability conditions set out in Section 3.2 below, where the target of the end-user is household and the ICS deployed is at least 25% of thermal efficiency.
2	Use the technologies or measures specified in the project description.	The technology used for the project activity is energy-efficient cookstoves. Only energy efficient cook stoves are to be adopted in the project by replacing traditional cookstoves in households.
3	Apply the technologies or measures in the same manner as specified in the project description.	Only energy efficient cook stoves are to be adopted in the project by replacing traditional cook stoves in households and same will be provided at free of cost.
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	The new project activity instances will be installed within Ghana subject to the same baseline scenario determined in Section 3.4 below.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All new project activity instances will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no mandated government programme or policy in the host country of this project ensuring

		the distribution of new energy efficient cook stoves for each project activity instance. Step 2: Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the applicability conditions of the methodology because the ICS will be distributed free of cost to the end users and project has no other source of revenue other than the sale of GHG credits.
6	Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: Each project activity instance that exceeds one percent of the capacity limit shall be identified. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit. (Ref. section 3.6.8 of the VCS standard v4.4)	No project activity instance shall exceed the applicable limit, which is 180 GWh_{th}/y. Expected annual energy saving for each instance comes out to be 0.0191 GWh_{th}/y which is less than 0.02% of the threshold limit. As the annual energy saving is below 1% of the limit, therefore no project activity instance is identified and divided into clusters.

Along with the above points, the Inclusion of New Project Activity Instances follows the below criteria

Grouped projects provide for the inclusion of new project activity instances subsequent to the initial validation of the project.

No.	Criterion	How the new project activity instances comply
1	Occur within one of the designated geographic areas specified in the project description	New project activity instances (Energy Efficient Cook Stoves) will occur in the designated geographic areas specified in the project description that is Ghana.
2	Conform with at least one complete set of eligibility criteria for the	New project activity instances will comply with a complete set of eligibility criteria as mentioned

No.	Criterion	How the new project activity instances comply
	inclusion of new project activity instances. Partial conformance with multiple sets of eligibility criteria is insufficient.	above for the inclusion of new project activity instances.
3	Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	New project activity instances information to be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body
4	Be included in an updated project description, with updated project location information (as set out in Section 3.11), which shall be validated at the time of verification against the applicable set of eligibility criteria.	New project activity instances should be included in project description section of respective Monitoring Report by validating eligibility criteria as mentioned in this document.
5	Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	New project activity instances ownership needs to be evidenced from the start date of the respective project activity instance.
6	Have a start date that is the same as or later than the grouped project start date.	New project activity instances' start date should be after the start date of grouped project activity.
7	Be eligible for crediting from the start date of the project activity instance through to the end of the project crediting period (only).	New project activity instances are eligible to claim credits from the start date of project activity to the end of the project crediting period.
8	Only eligible for crediting from the start of the verification period in which they were added to the grouped project.	New project activity instances will adhere to this criterion by being eligible for crediting only from the beginning of the verification period in which they are included in the grouped project

No.	Criterion	How the new project activity instances comply
9	Not be or have been enrolled in another VCS project.	New project activity instances will not be previously registered or currently enrolled in any other VCS project.
10	Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 - 3.6.9.	In line with section 3.6.8 of the VCS standard v4.4 all singular project activity instances within ten kilometres of another instance of this project activity with EKI Energy Services Limited will be consider within this current project activity only (Project ID 3801). Applicability for Section 3.6.9, already provided in the above table.

1.5 Project Proponent

Organization name	EKI Energy Services Limited
Contact person	Manish Dabkara
Title	Managing Director & Chief Executive Officer
Address	Office No 201, Plot No 48, Scheme 78, Vijay Nagar Part- II, Indore 452010, India
Telephone	+91 99075 34900
Email	manish@enkingint.org and registry@enkingint.org

1.6 Other Entities Involved in the Project

At present, EKI Energy Services Limited is the only sole entity of this project.

Organization name	EKI Energy Services Limited
Role in the project	Manish Dabkara
Contact person	Managing Director & Chief Executive Officer
Title	Office No 201, Plot No 48, Scheme 78, Vijay Nagar Part- II, Indore 452010, India

Address	+91 99075 34900
Telephone	manish@enkingint.org and registry@enkingint.org
Email	EKI Energy Services Limited

1.7 Ownership

The project ownership is with EKI Energy Services Limited.

A pre-survey was conducted to identify the communities, using families who only used traditional stoves rather than LPG or any other type of ICS. The ICS is distributed to end users (Households) free of cost. The end users are informed in advance that the use of ICS generates carbon finance which in turn will be used to cover the price of ICS and for recovering project implementation costs. The participating household joined the initiative voluntarily and willingly sign a declaration to transfer the ownership rights of the carbon assets generated from this project to EKI Energy Services Limited. The detail regarding end user agreement is explained clearly to the household community. As the ICS is distributed free of cost and therefore all the households are happy with the distribution. End users are informed that their participation in this project is voluntary and the same has been mentioned in the End User Agreement also, signed by the end user. The undertaking from the manufacturer and distributor and end-user agreement will be submitted to VVB to confirm that cook stove ownership is with EKI Energy Services Limited.

1.8 Project Start Date

16-January-2023 (The earliest date of commissioning of 1st ICS)

1.9 Project Crediting Period

First crediting period: 16-January-2023 to 15-January-2030 (inclusive of both the dates), seven years renewable crediting period. Being a renewable crediting period, the total crediting period of grouped project activity is 21 years.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	X
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 2023-24	77,417
Year 2024-25	75,591
Year 2025-26	73,784
Year 2026-27	71,994
Year 2027-28	70,223
Year 2028-29	68,469
Year 2029-30	66,732
Total estimated ERs	504,210
Total number of crediting years	7
Average annual ERs	72,030

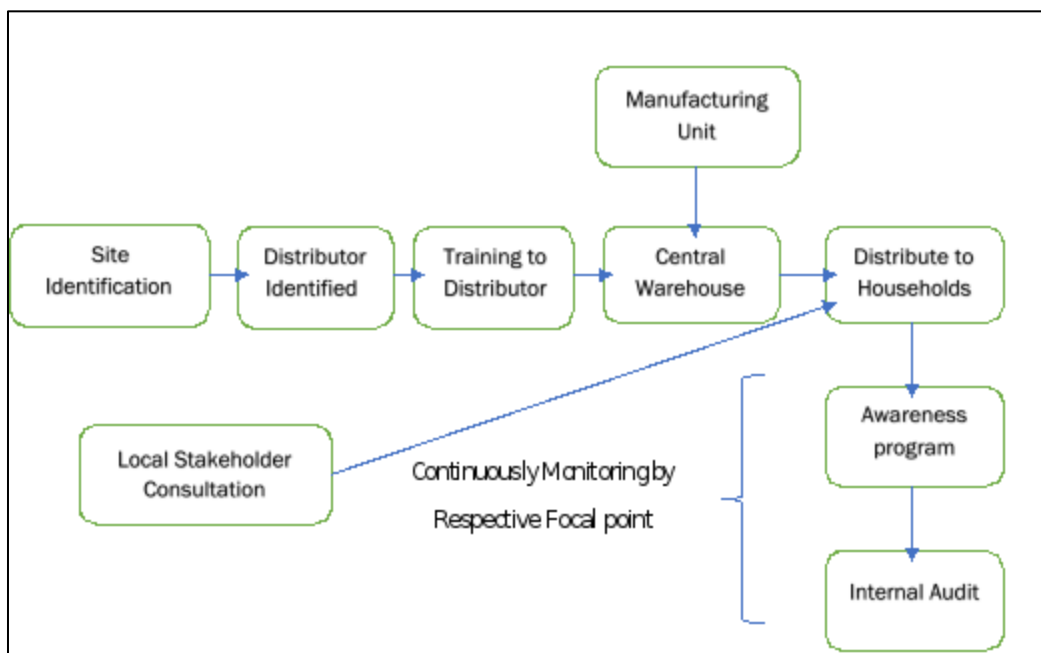
1.11 Description of the Project Activity

The project involves the distribution of fuel-efficient improved cookstoves (ICS) to replace the baseline traditional cookstoves in households.

The ICS deployed under this project is energy efficient which substantially reduces fuel consumption (charcoal) and emissions for conducting cooking and water heating tasks in homes. The ICS improves the efficiency of combustion and thermal transfer to the pot compared with traditional pot support or three-stone fire by incorporating energy-efficient technology which provides a conducive environment for clean and efficient combustion of charcoal. It substantially reduces charcoal fuel consumption compared with a less efficient three-stone fire or traditional pot support.

EKI has procured the ICSs from the manufacturer and provided the same to the distribution agency at a central point near the distribution area. The distribution agency has trained its teams for the distribution and registration process. Further, the distribution team have distributed the stoves to the end users. The distribution team have educated the end-user on the proper use of the ICS, its benefits for the user, and how it improves the environment. Additionally, EKI is also

planning to conduct regular training programs and field visits for ensuring continual communication with the local stakeholders. A field visit was undertaken as part of the initial project activity instances. Furthermore, in case of any support for repair/maintenance of the stove, the end-user can contact the support team using the customer support number available on the box of ICS.



Cookstove Flow Chart

For the implementation of the improved stoves, EKI has identified the potential location where the majority of the population was using the traditional mud and/or three-stone charcoal fire stoves. Based on available information, or baseline survey, ICS implementation was planned. EKI prioritized the appointment of a local implementation partner for the delivery of ICS and local men's and women's teams contracted and trained for the implementation of ICS. This boosts local employment and the income of local people. Awareness about the implementation and use of the ICS have been provided to the local community through awareness programs. EKI has conducted the distribution of ICS door-to-door or by selecting a dedicated central point accessible to the intended user. EKI has deployed a dedicated team to train end users for the proper use of ICS and to educate users about the benefits towards climate with the use of ICS. The team ensures the proper functioning of ICS and take feedback from users about ICS and address their concerns.

The ICS deployed under the grouped project activity may be either the portable or fixed cookstove manufactured by the different manufacturers, having thermal efficiency of includes more than 25% (as per applied methodology VMR 0006).

Technical Specifications of cookstoves distributed under the project activity instance to date are as below:

TECHNICAL DETAILS		
A)	Cook Stove Type/Category	Gyapa 
B)	Fuel Used	Charcoal
C)	Weight	8 Kg
D)	Wide	290 mm
E)	Height	260 mm
F)	Body Material	AluZinc Sheet
G)	Material Thickness	0.5 mm
H)	Air Supply	Natural Draft
I)	Expected life of the stove	7 Years
J)	Guarantee /Warranty Period	1 Year
L)	The Overall Thermal Efficiency	37%

TECHNICAL DETAILS		
A)	Cook Stove Type/Category	Obaapa 

B)	Outer Body Material	Aluzinc Sheet
C)	Inner Chamber Material	Ceramic
D)	Material of Handle	Combustion Chamber
E)	Material of Wood Support	N/A
F)	Material of Led Provided	Iron Rod Pot Support
G)	Length of Stove (in mm)	260
H)	Width of Stove (in mm)	290
I)	Expected life of the stove	7 Years
J)	Guarantee /Warranty Period	1 Year
K)	Weight	Above 6 kg
L)	The Overall Thermal Efficiency	36%

The above specifications have been considered for actual ER calculation of the project activity instances in this Joint PD & MR.

The PP may change for future models of ICS or add different models of ICS and the same will be transparently reflected during verification and the details and specifications of respective cook stoves will be considered for ER calculation of project activity instance.

The thermal efficiency of the implemented cookstove was tested at the manufacturer's end. Further, continuous research and development activities are being conducted by the manufacturer to improve the efficiency, quality, and design of the stove, by incorporating feedback from end users through routine follow-ups and site visits. EKI may adopt the modified stove models with improves efficiency for future distribution under the current grouped project activity.

Data collection of ICS end-user

ICS distribution agency is responsible for stove distribution and data collection. The distribution agency follows a SOP designed by EKI for stove distribution and registration. The ICS registration process was carried out by manual recording process or using ICT¹ (e.g., using a mobile app). The distribution team gets training that includes a brief introduction to the project activity, the objective of the distribution of ICS as well as ICS distribution and data collection process.

¹ Information Communications Technology

The project proponent gathered the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS have been assigned with a unique serial number. This number has been recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address/ Village name of ICS household
- Stove model Distributed
- Date of distribution/installation
- ICS serial number
- Retailer/distributor/Manufacturer information

The information collected have been stored in hard copies and/or electronic database excel sheets which serves as a project database for project monitoring and sampling purposes.

1.12 Project Location

All the project activity instances in the proposed grouped project activity have been located within the geographical boundaries of Africa.

For the initial project activity instance, the geographical area is in Republic of Ghana though it may expand in different countries of Africa.

The geographical boundary for projects located in Republic of Ghana is delineated in the form of extreme geographic coordinates of Ghana as follows²:

Latitude: 8° 2' 5.5608" N

Longitude: - 1° 18' 47.8512" W

²

<https://www.google.com/maps/place/Ghana/@0.5907892,11.6986528,3z/data=!4m5!3m4!1s0xfd75acda8dad6c7:0x54d7f230d093d236!8m2!3d7.946527!4d-1.023194>



1.13 Conditions Prior to Project Initiation

The conditions prior to project initiation are the continued use of non-renewable fuel (charcoal) by the target population to meet similar thermal energy needs as provided by project cook stoves in absence of project activity.

A large number of households in rural Ghana rely on mud or three-stone stoves for their daily cooking needs. These mud or three-stone stoves not only have very low thermal efficiency (20% as per the applicable methodology) but also contribute to health hazards from indoor air (smoke) pollution.

Through the current project activity, the distribution and installation of highly thermal efficient improved cook stoves (ICS) are undertaken to replace the inefficient traditional or three stone fire cookstoves. The ICS burns charcoal more efficiently thereby improving thermal energy transfer to pots, hence saving charcoal and greenhouse gases.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of improved cookstoves in Ghana for households. The project is a voluntary effort by the project proponent.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading. The undertaking is submitted for the same.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit. The undertaking is submitted for the same.

1.16.3 Supply Chain (Scope 3) Emissions

There are no measurable Scope 3 emissions associated with the grouped project activity.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project includes the distribution of fuel-efficient improved cookstoves thereby replacing the use of inefficient cookstoves. The project contributes to social, environmental, economic, and technological benefits which contribute to the sustainable development of the local environment and the country as follows:

Social benefits

1. Reduces drudgery of women (due to reduced charcoal use) who spend long hours and travel long distances to collect charcoal. Hence reducing the proportion of time spent on unpaid domestic and care work (SDG 5.4.1)
2. Improves the overall health of women and children by reducing smoke in the kitchen, thus reducing health hazards from indoor air pollution. Therefore, contribute to a

decrease in the mortality rate attributed to household and ambient air pollution (SDG 3.9)

3. Better cooking time – the materials used in making the ICS transmit the heat effectively, cooking the food faster. Hence reducing the proportion of time spent on unpaid domestic and care work (SDG 5.4.1)
4. Better cooking environment due to less smoke and carbon residue in the kitchen. Therefore, contribute to an increase in the proportion of the population with primary reliance on clean fuels and technology (SDG 7.1.2)
5. Better quality of life – the rural communities get family time as the whole family can sit and eat together. As the cooking on ICS is faster due to its thermal energy-based mechanism. Therefore, this project can improve the quality of life of the end-user.

Environmental benefits

1. Improves the local environment by reducing the rate of degradation of forests and deforestation in the project area by reducing charcoal consumption. (SDG 15.2.1)
2. Reduce indoor pollution – ICS emits less smoke and reduces morbidity from respiratory diseases and other health hazards.
3. Reduce global and local environmental pollution and environmental degradation by reducing in use of non-renewable biomass thus leading to a reduction in GHG emissions (SDG 13.0).

Economic benefits

1. Employment opportunities for local communities through the VCS activity – e.g. employments related to Manufacturing of ICS, Distribution and Monitoring activity (SDG 8.3).
2. Reduces purchase of charcoal and/or wage equivalent from reduced charcoal collection time.
3. A resource-poor household would need to spend limited available finances on medicines, further exacerbated by the loss of wages from both not being able to work and having to look after the ill person. Due to reduced indoor pollution, a reduction in morbidity from respiratory diseases and other health hazards, as well as the medical expenditure involved will be witnessed. Therefore, it will contribute to reducing the poverty of the participating households (SDG 1.1.1).

Technological benefits

1. Implementation of ICS will contribute to “Proportion of population with primary reliance on clean fuels and technology” by introducing efficient technology to the rural communities (SDG 7.1.2).

1.17.2 Sustainable Development Contributions Activity Monitoring

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
<i>Sequential row number</i>	<i>SDG Target number</i>	<i>Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator</i>	<i>Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)</i>	<i>Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.</i>	<i>Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.</i>
1)	1.1	1.1.1 Proportion of the population living below the international poverty line by sex, age, employment status and geographic location (urban/rural)	Implemented activities to decrease	The project has increased the 80 participants' total daily income to GHS 2000 to 5500 per month, bringing them above the international poverty line	The project has increased the 80 participants' total daily income to GHS 2000 to 5500 per month, bringing them above the international poverty line

2)	3.9	Project-specific indicator: Number of households experiencing reduction in particulate matter.	Implemented activities to decrease	Reduce the fine particulate matter (PM2.5) in 20,102 households emitted during cooking thereby improving overall health outcomes and reducing burdens of disease.	Reduce the fine particulate matter (PM2.5) in 20,102 households emitted during cooking thereby improving overall health outcomes and reducing burdens of disease.
3)	7.1	7.1.2 Proportion of population with primary reliance on clean fuels and technology	Implemented activities to increase	Increase access to clean cooking technology with ICS installations in 20,102 households under the project activity	Increase access to clean cooking technology with ICS installations in 20,102 households under the project activity
4)	8.3	Project specific indicator: Total Number of jobs created.	Implemented activities to increase	Project activity created a total of 93 new job positions in different roles for manufacturing and distribution of ICS and post-implementation monitoring activity.	The project activity has created job opportunities for 93 heads (Male:Female::58:35) in the ICS manufacturing unit
5)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to decrease	Contribute to greenhouse gas emission reduction of 4,291 tCO ₂ e in current monitoring period (16-January-2023 to 28-February-2023)	Prevented the release of 4,291 tCO ₂ e into the atmosphere in ICSs lifetime.

6)	15.2	15.3.1 Proportion of land that is degraded over total land area	Implemented activities to decrease	Contribute to the reduction of deforestation of 2,555 tonnes of charcoal biomass from forests surrounding the communities and reducing pressure on forest reserves	Reduced deforestation of 2,555 tonnes of charcoal biomass from forests surrounding the communities and reduced pressure on forest reserves
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1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

The project only brings positive impacts on environmental and socio-economic aspects as elaborated in Section 1.17 above.

The project contributes to social, environmental, economic, and technological benefits which contribute to the sustainable development of the local environment and the country such as better cooking time & environment, reduction in indoor pollution, employment opportunities, and many more.

No potential negative environmental or socio-economic impacts have been identified for the project; thus, this section is not applicable.

2.2 Local Stakeholder Consultation

Identification of households was done based on a general survey. A baseline scenario was calculated by assessing the number of households using mud or three-stone stoves and the potential for emission reductions by replacement of mud or three-stone stoves by ICS was checked.

Various awareness activities such as pamphlet distribution, door-to-door awareness, invitations, etc were conducted to engage community members and invite participants in the local stakeholder meetings and the project.

The villages were identified and village representatives like village heads were invited. For the proposed project activity, household owners (end users), village heads, distributors, and their representatives are relevant main local stakeholders. The public notice was placed in common

places of villages and village heads were invited telephonically and personally to consult their feedback.

Further, PP has also planned to conduct an awareness program/meeting in the new location within the geographical area to the local community for proper use, benefits of the cookstove, and benefits towards the environment. EKI has conducted these meetings where the need for such an awareness program were required. Such meetings were also named local stakeholder consultation meetings. Details of the same are also provided below:

The details of the Stakeholder Meetings are as follows:

Date of invitation – 27-August-2022

Date of Meeting – 15-October-2022

Location of Meeting – Miklin Hotel, Accra, Ghana

Following is a brief description of the meetings conducted to date:

In the introductory speech, the representative of EKI Energy Services Limited (Project Investor), welcomed the gathering and gave a brief about the climate mitigation project activity. After the introductory speech, stakeholders were explained about the distribution of highly efficient cook stoves which will burn charcoal more efficiently thereby improving thermal transfer to pots, hence saving fuel and time. The project also reduces the rapidly progressing deforestation but also reduce health hazards from indoor smoke pollution and women and children now have to spend less time collecting contributing to a reduction in GHG emissions. The Project also assist the local population by providing employment opportunities to both skilled & unskilled labours.

The Minutes of the meetings with stakeholders' queries, and invitation letters will be submitted to the VVB.

The meetings started with an opening speech by a representative of the project participant. He introduced all guests to the dais. The representative of the project participant explained the technical aspects of the project to the stakeholders. He also explained about social, environmental & economic benefits of the project. He also elaborated about the carbon mechanism & its requirement for the current project. After the detailed discussions, the session was open to questions from stakeholders. Most of the questions were related to employment opportunities, economic development, benefits from the project to villagers, and other development activities.

Some of the question raised by the villagers in **Ghana** are summarized below:

1. District – Accra

Name of Person: Frank Sekyere

Q: How Sustainable is this initiative?

A: The project involves sensitisation on continue usage and switch to ICS from inefficient stove. Carbon financing makes this possible.

Name of Person: Asantewa Agyekum

Q: How will you ensure the stoves reaches the rural areas?

A: We are using local agents with on-ground connection to community that he /she also resides. Our distribution is house to house in rural areas.

Name of Person: Sarfoh Danguah

Q: How can someone be a distributor for the clean cookstoves?

A: You can be a distributor not for sale of stoves but for community house to house distribution.

Name of Person: Deladem Kojo Xam

Q: What available options are for organisation that manufacture briquettes?

A: we will be interested to explore briquettes if only the supply chain to ensure fuel availability within community can be guaranteed after an assessment.

Name of Person: Ivy Ewurama Adams

Q: How durable is this stove, because the usual clean cook stoves of the market easily break as a result of the local foods eg: (Banku, Konkontey, eworkney) we cook in Ghana. Are they durable enough to assist us cooking?

A: Durability of this Project is guaranteed through warranty.

Name of Person: Ernard Aihon

Q: What are your plant towards improving the features and qualities of the stove.

A: Our follow-up on stoves in housefold will informs us to explore customer inputs/suggestions for improvement.

Name of Person: Mr. Ampomach Isaac (Concern Health)

Q: Is there stove and manual available for stove users?

A: Yes, we will insert a manual with pictogram/ graphic demonstration for users.

Name of Person: Yahaya Yakubu

Q: Are the available sizes/ bigger ones for commercial users.?

A: The project target household hence have medium size.

Name of Person: Fanny Aba Appiah

Q: How will you ensure stove guaranty is guaranteed and meet customer satisfaction?

A: We provide one year warranty, a contact to lodge complains and also follow-ups

Name of Person: Emnancsella Doreen Kwofie

Q: How do you maintain the stoves to guarantee long time use?

A: Avoid using stove in rain or avoid quenching fire in ceramic chamber when hot. Do not drop stove from height.

Name of Person: Salomey Adjei

Q: Is it applicable to use clay for maintenance?

A: This is no need for indigenous maintenance of mixing clay and wetting stove for maintenance.

Name of Person: Daniella Yeboah

Q: How safe is stove. I hope the stove body does not get hot?

A: The stove safety does not get too hot but precaution should be taken since fire is involved.

The stakeholders are satisfied due to the health and environmental benefits and there are no negative comments from stakeholders.

As part of PP's ongoing communication efforts, PP provided end users with the contact information of the on-ground team members deployed in their respective localities. The on-ground team shall diligently report any grievances received from the end users on a daily basis to the distributors they are accountable to. These distributors shall then document all the grievances in a dedicated register and provide regular updates to our PP representative. PP maintains regular communication with the distributor to ensure that all grievances are promptly addressed on an urgent basis. This process reflects PP's commitment to maintaining continuous stakeholder communication through an effective grievance mechanism. There were no grievances were received during the current monitoring period.

EKI is also planned to contact the end users on regular basis to ensure the proper functioning of cookstoves, take feedback to address grievances, and add more users by spreading awareness. EKI team is planned to visit the site selection random location for this purpose and mainly record the information related to regular usage of ICS, the physical condition of ICS, benefits observed by the usage of ICS, information related to the usage of baseline cook stoves, etc.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project. In fact, project activity supports various positive environmental impacts such as -

1. Improves the local environment by reducing the rate of degradation of forests and deforestation in the project area.
2. Reduce indoor pollution – ICS emits less smoke and reduces morbidity from respiratory diseases and other health hazards.
3. Reduce global and local environmental pollution and environmental degradation by a reduction in the use of non-renewable biomass thus leading to a reduction in GHG emissions.
4. Less water and effort are needed for cleaning vessels as the cooking process is relatively smoke-free.

2.4 Public Comments

Between 16-January-2023 and 15-February-2023, the public comment period, no comments were received for the project activity.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

VCS Methodology: VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1 dated 22 July 2021, Sectoral scope 3

<https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

This methodology also refers to the latest version of AMS II.G version 13 - Energy efficiency measures in thermal applications of non-renewable biomass

<https://cdm.unfccc.int/methodologies/DB/GNFWB3Y6GM4WPXFRR2SXKS9XR908IO>

For the calculation of the fraction of non-renewable biomass, the below tool is used “TOOL30 version 4.0: Calculation of the fraction of non-renewable biomass”

<https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-30-v4.0.pdf>

3.2 Applicability of Methodology

The applicability of the methodology is justified below

Applicability criterion	How the project complies
Project activities shall be implemented in domestic premises or community-based kitchen	The proposed project involves the deployment of ICS only in households. For all project activity instances, project activity is implemented in households.
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single-pot or multi-pot;	Energy Efficient stoves planned to be installed under this project are single-pot portable charcoal fired cook stoves that have an efficiency of more than 25%. For all project activity instances, ICS with an efficiency of 37% and 36% respectively is planned to be installed. For future project activity instances, the manufacturer may change or add, and the cook stove efficiency

	that should be more than 25% as mentioned in the methodology based on manufacturer specification.
Both 'Projects' and 'Large Projects' can use the methodology	Estimated average annual emission reductions for the grouped project activity are less than 300,000 tonnes of CO ₂ e per year. Therefore, the proposed project qualifies for the "Projects" criteria.
Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports, or statistics;	Non-renewable biomass has been used since 31 December 1989 as demonstrated by UN Data ³ and as demonstrated below by referring to published literature, official reports, or statistics;
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: It is produced using exclusively renewable biomass (more than one type of biomass may be used) The consumption of the fuel should be monitored during the crediting period Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting)	Not applicable, as the project activity does not use processed biomass as a fuel. The ICS is introduced as an energy efficiency measure to replace baseline stoves and reduce the use of non-renewable biomass for combustion.
The VCS PD shall explain the proposed method for the distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo) The above criteria are as per the below VCS meth requirement and para 7 of the latest version of meth AMS II.G version 12 is followed. (Additionally, applicability criteria numbers 8 and 9 set out in Section 2.2 of AMS II.G, version 11.1 shall apply)	Each ICS in this project can be identified by a unique combination of customer name and geographical location, as well as a serial number. The serial number is a unique number which allows for a clear distinction between the stoves. No individual serial number can be repeated within the project, thus ensuring that each stove is counted only once in the proposed project. In addition, the project has been cross-checked against other CDM project activity operating in the country using the UNFCCC, the Gold Standard, and other relevant voluntary carbon schemes to ensure that the ICS is not included in any other CDM project activity or voluntary project activity.
The VCS PD shall also explain how the proposed procedures prevent double counting of emission reductions, for example, to avoid the project stove manufacturer, wholesale providers, or others claiming credit for emission reductions from the project devices. The above criteria are as per the below VCS meth requirement and para 8 of the latest version of meth AMS II.G	Manufacturers/ cook stove distributors undertaking is provided that EKI is the owner of that carbon credits and Manufacturers/ cook stove distributors will not claim any credits for such cook stoves.

³ UNdata | record view | Fuelwood

version 12 is followed. (Additionally, applicability criteria numbers 8 and 9 set out in Section 2.2 of AMS II.G, version 11.1 shall apply)	
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Applicability criterion	How the project complies
This tool may be used by: (a) DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or (b) project participants to calculate project- or PoA-specific f_{NRB} values	The project participant has used this Tool to calculate project-specific f_{NRB} values.

Project participants can show that non-renewable biomass has been used since 31 December 1989, using survey methods or referring to published literature, official reports, or statistics.

Justification: The use of non-renewable biomass is established through a demonstration of the gap between the demand for firewood and silviculturally permissible production of biomass from forests and trees outside the forest. The fact is established using published literature referring to the following studies:

According to a Ghana's national report (2017)⁴, the rate of deforestation and forest degradation has been on the rise in recent decades. The annual deforestation rose to 794,214 ha per annum between 2013 and 2015. Much has been written about deforestation's causes in Ghana, and it is reported (2019) that Non-renewable biomass such as firewood and charcoal is a key driver of deforestation and negatively impacts climate change, biodiversity, and livelihoods in Ghana⁵. Thus, the above clearly shows that non-renewable biomass has been used in the Ghana since 31 December 1989.

Conclusion:

Therefore, it is established from the above studies that, a staggering gap exists between the demand and potential for sustainably extractable fuel wood from forest land. While the first three studies refer to the fuel wood supply scenario and the gap that were precedent before 1989; the study highlight the widening of gap between the demand and supply of biomass after 1989. The widening of the gap has been both due to the increased population pressure as well as the conversion of dense forest to medium dense, scrub and open type forest land due to

⁴ <https://www.state.gov/reports/2017-country-reports-on-human-rights-practices/ghana/>

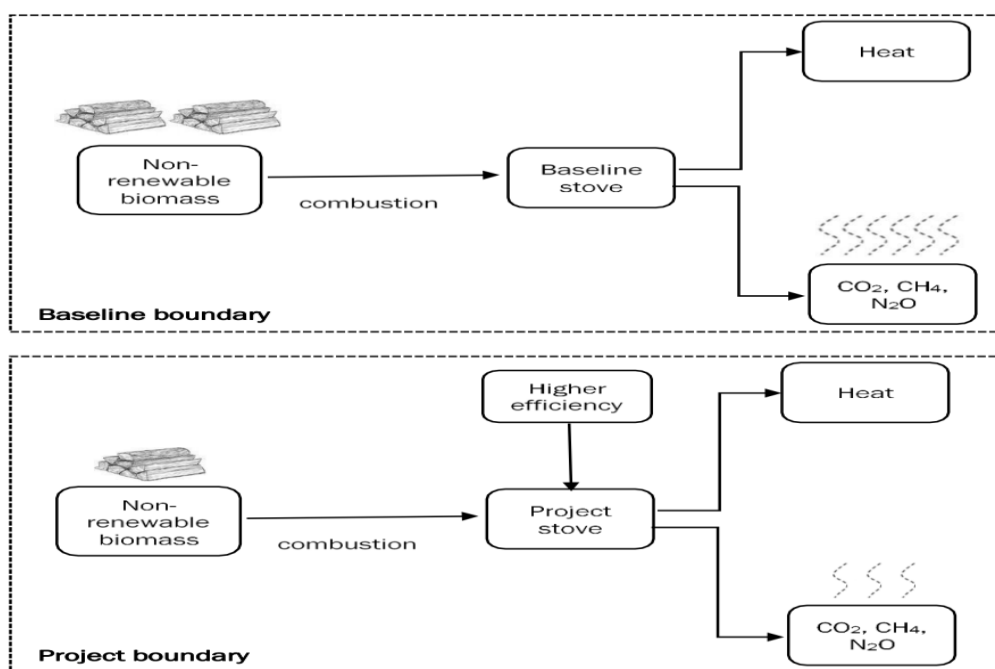
⁵ D.Nje et al., Greening The Wood Fuel Sector Of Ghana, (2019), https://collections.unu.edu/eserv/UNU:7280/UNU-INRA_PB_STEWART-UNUIGBE_FEB2019.pdf

unsustainable extraction even after the government's initiative towards promotion of afforestation. Thus, it can therefore be concluded that the gap exists from and before 1989 and continued thereafter, forcing the population in using biomass extracted in un- sustainable manner which is identified as non-renewable component

3.3 Project Boundary

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Major source
	CH ₄	Yes	Major source
	N ₂ O	Yes	Major source
	Other	No	No other source identified
Project	CO ₂	Yes	Major source
	CH ₄	Yes	Major source
	N ₂ O	Yes	Major source
	Other	No	No other source identified

A representation of the baseline boundary and project boundary is given below



3.4 Baseline Scenario

As per methodology, VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1, Section 6 baseline scenario “The baseline scenario is the continued use of non-renewable wood fuel (firewood/charcoal) or fossil fuel (coal/kerosene) by the target population to meet similar thermal energy needs as provided by project cook stoves in absence of project activity.”

The project activity instances to be included in this grouped project activity should meet the baseline scenario as mentioned above in section 1.4. For the project activity, the baseline scenario is the continued use of non-renewable fuel (charcoal) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

With the baseline survey result it was concluded that the baseline for this project is the ongoing use of non-renewable wood fuel in traditional cookstoves or three-stone fire stove within a specific region.

3.5 Additionality

The methodology uses the activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in the host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute, or another regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or another regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by EKI.

Step 2: Positive List

As per Section 3.2, the project meets the applicability conditions of the methodology which represent the positive list.

The project installs the ICS at zero cost to the household and has no other source of revenue other than the sale of GHG credits.

The project is not implemented as part of government schemes or supported by multilateral funds.

Conclusion: As the project fulfills the conditions above, it is deemed additional.

If any project activity instance does not follow the positive list as stipulated by methodology, then additionality needs to demonstrate as per tool requirement.

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

Till the end of the monitoring period, 20,102 project activity instances (ICS) were deployed under this grouped project. Only two stove model (Gyapa cookstoves and Obaapa Clean Stove) was distributed in the project.

VCS methodology VMR0006, version 1.1 allows the use of a correction factor of 0.95 applied to the overall emissions reductions to account for any possible leakage. This factor has been applied to the emission reductions presented in this report.

During the current monitoring period, no incident or event occurred, that could affect GHG emissions reduction and the approved monitoring plan.

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The methodology does not account for baseline emissions separately but instead quantifies net emission reductions achieved by the project. Please refer to Section 5.4.

5.2 Project Emissions

The methodology does not account for project emissions separately but instead quantifies net emission reductions achieved by the project. Please refer to Section 5.4.

5.3 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology.

5.4 Estimated Net GHG Emission Reductions and Removals

The improved cookstove is introduced as an energy efficiency measure in the project, therefore equations 1 and 2 of the methodology is applied to calculate the net GHG emission reductions

$$ER_y = \sum_i \sum_j ER_{y,i,j}$$

Where:

- i = Indices for the situation where more than one type/model of improved cookstove is introduced to replace three-stone fire
- J = Indices for the situation where there is more than one batch of improved cookstove of type i
- ER_y = Emission reductions during year y in tCO₂e
- $ER_{y,i,j}$ = Emission reductions by improved cookstove of type i and batch j during year y in tCO₂e

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95 \quad \text{Equation (2)}$$

Where:

- $B_{y,savings,i,j}$ = Quantity of woody biomass (Charcoal) that is saved in tonnes per improved cookstove of type i and batch j during year y
- $f_{NRB,y}$ = Fraction of woody biomass that can be established as non-renewable biomass (f_{NRB})⁶
- $NCV_{wood\ fuel}$ = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne)⁷
- EF_{wf,CO_2} = CO₂ emission factor for the use of wood fuel in the baseline scenario (IPCC default for wood fuel, 112 tCO₂/TJ)⁸
- $EF_{wf,non\ CO_2}$ = Non-CO₂ emission factor for the use of wood fuel in the baseline scenario (IPCC default for wood fuel, 26.23 tCO₂/TJ)⁹
- $N_{y,i,j}$ = Number of improved cookstoves of type i and batch j operating during year y

⁶ Default values endorsed by designated national authorities and approved by the Board are available at <https://cdm.unfccc.int/DNA/fNRB/index.html>

⁷2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction

⁸2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

⁹2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

0.95 = Discount factor to account for leakage

The quantification of charcoal saved due to the implementation of improved cookstoves to be estimated using equation below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right) \quad \text{Equation (3)}$$

Where

- η_{old} = Efficiency of baseline cookstove
- $\eta_{new,y,i,j}$ = Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y . Alternatively, efficiency may be determined using Equation 4.
- $B_{y=1,new,i,j,survey}$ = Annual quantity of charcoal used by improved cookstoves in tonnes per device of type i and batch j , determined in the first year of the implementation of the project through a sample survey.

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad \text{Equation (4)}$$

Where

- η_p = Efficiency of project stove (fraction) at the start of project activity.
- $(DF_n)^{y-1}$ = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively, default value of 0.99 efficiency loss per year can be considered.
- 0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

For ex-ante calculation purpose, the assumption below is applied:

- 1) The project installed 20,102 ICSs.
- 2) The life span of each ICS is 7 years; thus, the operational lifetime of each project activity instance is taken as 7 years.

- 3) Annual stove loss rate¹⁰ is estimated at 0%. This is assumed for estimation. During actual ER calculation, this value may change. This value are determined during actual verification based on survey results.
- 4) $B_{y=1, \text{ new, i, survey,}}$ is considered as 8.55 kg/device/day or equal to 3.119 tonnes/device/year for Gyapa. And $B_{y=1, \text{ new, i, survey,}}$ is considered as 8.36 kg/device/day or equal to 3.051 tonnes/device/year for Obaapa.

This value has been determined during the first monitoring period, based on survey results.

Determination of F_{nr} value:

The fraction of woody biomass that can be established as non-renewable is given by:

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Where:

f_{NRB}	=	Fraction of non-renewable biomass in the applicable area in the relevant period (fraction or %)
NRB	=	Quantity of non-renewable biomass consumed in the applicable area in the relevant period (tonnes)
RB	=	Quantity of renewable biomass that is available on a sustainable basis in the applicable area in the relevant period (tonnes)

Commercial use of woody biomass for non-energy purposes (such as building or furniture) that is derived from forests or other land areas in the relevant area (tonnes)

$$NRB = H - RB$$

Where:

H	=	Total consumption of woody biomass in the applicable area in the relevant period (tonnes)
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The following equation accounts for all consumption within the relevant area to determine the overall consumption of woody biomass (H).

$$H = HW \times N + CE + NE$$

¹⁰ The "Annual stove loss rate" in the context of Improved Cook Stoves (ICS) refers to the annual rate at which ICS units are lost or removed from service, typically due to wear and tear, damage, or other reasons. It represents the number of ICS units that become non-operational or are taken out of use in a given year.

Where:

- HW = Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes//household)
- N = Number of households consuming wood fuel within the applicable area in the relevant period (number)
- CE = Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)
- NE = Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

The total consumption of woody biomass in the applicable area has been calculated from the direct available data for Total roundwood consumption and wood density in the following table.

Parameter	Value	Unit
Total roundwood consumption ¹¹	52.17	In Million cums
Wood density ¹²	0.725	T/m3
H (Total consumption of woody biomass in the applicable area in the relevant period)	37.82	Million Tonnes

Procedure to estimate the quantity of renewable biomass available (RB):

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i}))$$

Where:

- $MAI_{forest,i}$ = Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas in the relevant period (tonnes/ha/yr)
- $MAI_{other,i}$ = Mean Annual Increment of woody biomass growth per hectare in sub-category i of other land areas in the relevant period (tonnes/ha/yr)
- $F_{forest,i}$ = Extent of forest in sub-category i in the relevant period (ha)
- $F_{other,i}$ = Extent of other land in sub-category i in the relevant period (ha)
- $P_{forest,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in sub-category i) in the relevant period (ha)

¹¹ FAO Yearbook of Forest Products 2020 (page 82)

¹² THE CHARCOAL TRANSITION (Page 142, TABLE A7 Typical properties of dry wood, fuelwood and charcoal)

$P_{other,i}$ = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in sub-category i) in the relevant period (ha)

Distribution of total forest area (Ghana) by ecological zone¹³:

Country	Tropical					
	Sudan Savannah	Guinea Savannah	Deciduous forest	Semi-Deciduous	High Rainforest	Costal Savannah
Ghana	1%	63%	3%	28%	3%	2%
Ghana Rainfall (mm/yr)	1000	1100	1500	1300	2200	800
Tropical Zone	Dry	Moist	Moist	Moist	Rain forest	Dry

Distribution of total forest area by age:

	Primary forest ¹⁴	Established area (forest ≤ 20 yrs) ¹⁵	Forest area ≥ 20yrs	Total forest
Area (million Ha)	0.39	0.27	7.34	7.99
	4.82%	3.32%	91.86%	100.00%

¹³ Understanding climate variability and change: analysis of temperature and rainfall across agroecological zones in Ghana, Page2, table 1

¹⁴ Global Forest Resources Assessment (FRA) 2020 Ghana - Report (fao.org), Page18

¹⁵ Forestry Commission Annual Report – 2020 – Forestry Commission (fcghana.org) & NATIONAL FOREST PLANTATION DEVELOPMENT PROGRAMME

~~Above-ground biomass growth (tonnes d.m./ha-yr)~~

~~Above-ground biomass growth(tonnes d.m./ha-yr)~~

[/PCC , Forest Land, Volume 4, Table-4.9](#)

Ecological Zone	Status Condition	Above Ground Biomass	Unit
Tropical rainforest	Primary	1.30	tonnes/hectare/year
	Secondary >20 years	3.50	tonnes/hectare/year
	Secondary ≤20 years	7.60	
Average		3.53	tonnes/hectare/year
Tropical moist deciduous forest	Primary	0.40	tonnes/hectare/year
	Secondary >20 years	0.90	tonnes/hectare/year
	Secondary ≤20 years	2.90	
Average		0.94	tonnes/hectare/year
Tropical Dry Forest	Primary	0.00	tonnes/hectare/year
	Secondary >20 years	1.60	tonnes/hectare/year
	Secondary ≤20 years	3.90	
Average		1.60	tonnes/hectare/year

Above-ground biomass growth:

$$MAI_{forest,i} = 1.04 \text{ tonnes/hectare/year}^{16}$$

$$MAI_{other,i} = 1.04 \text{ tonnes/hectare/year}$$

	$MAI_{forest,i}$	$MAI_{other,i}$	$F_{forest,i}$	$P_{forest,i}$	$F_{other,i}$	$P_{other,i}$	RB
	tonnes/hectare/year		Million Ha				Million tonnes
Ghana	1.04	1.04	7.99 ¹⁷	1.425 ¹⁸	0 ¹⁹	0 ²⁰	6.82

The fraction of woody biomass that can be established as non-renewable for all locations:

¹⁶ [IPCC, Forest Land, Volume 4, Table-4.9](#)

¹⁷ [Global Forest Resources Assessment 2020, Ghana. p.10 \[Forest \(a\)\]](#)

¹⁸ [Global Forest Resources Assessment 2020, Ghana. p.41 \[Forest area within protected areas\]](#)

¹⁹ Extent is unknown and is therefore reported as zero. Most wooded land in the Savanna are classified as forest as they meet Ghana's forest definition thresholds.

²⁰ Non-accessible area is reported as zero. All geographical areas are classified as protected or forest areas and are accessible.

	H	RB	NRB	f _{NRB}
Ghana	37.82	6.82	31.00	81.97%

In contrast, PP has rigorously employed the most recent available data, specifically from 2020, to compute an f_{NRB} value of 81.97% in accordance with Tool 30 version 4. This figure is anticipated to be more reliable compared to the provided literature and aligns more conservatively with the default value (99%) utilized during the period from 09 October 2013 to 08 October 2018.

The literature²¹ provided in paragraph 6. (b) (i) of Tool 30 version 4, published on 19 January 2015, presents a value range of 30% ~ 40% for f_{NRB} in Ghana (Figure 2). The literature explicitly states, "*We present a spatially explicit assessment of pan-tropical wood fuel supply and demand, calculate the degree to which wood fuel demand exceeds regrowth, and estimate wood fuel-related greenhouse-gas emissions for the year 2009.*" This underscores that all the values utilized in the report pertain to data preceding the year 2009.

Moreover, there is no explicit mention of the calculation methodology employed to derive the f_{NRB} values, which introduces uncertainty about its alignment with Tool 30 version 4. Additionally, this value (30% ~ 40%) is lower than the value (99%) published on 09 October 2013²² as Fraction of non-renewable biomass (f_{NRB})²³, which was universally employed by all projects until 08 October 2018.

Determination of efficiency of ICS during year y

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where

η_{p1} = 37% (Gyapa project stove)

η_{p2} = 36% (Obaapa Clean stove)

$$DF_n = 0.99$$

Example of calculation:

If project stove is Gyapa project stove and y= 1

$$\eta_{new,y,i,j} = 37\% \times (0.99)^{1-1} \times 0.94$$

34.78%

²¹ <https://www.nature.com/articles/nclimate2491>

²² <https://cdm.unfccc.int/DNA/fNRB/docs/ghana.pdf>

²³ <https://cdm.unfccc.int/DNA/fNRB/index.html>

Hence the efficiency of ICS during year y is as below:

Year (y)	$\eta_{new,y,i,j}$
1	34.78%
2	34.43%
3	34.09%
4	33.75%
5	33.41%
6	33.08%
7	32.74%

Example of calculation:

If project stove is Obaapa project stove and y= 1

$$\eta_{new,y,i,j} = 36\% \times (0.99)^{1-1} \times 0.94$$

33.84%

Hence the efficiency of ICS during year y is as below:

Year (y)	$\eta_{new,y,i,j}$
1	33.84%
2	33.50%
3	33.17%
4	32.83%
5	32.51%
6	32.18%
7	31.86%

Determination of the quantity of woody biomass that is saved in tonnes per ICS during year y

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right)$$

Example of calculation for Gyapa:

If y= 1,

$$B_{y,savings,i,j} = 3.119 \times [(0.3478/0.2) - 1]$$

= 2.31 tonnes

Year (y)	$B_{y=1,new,i,survey}$	$\eta_{new,y,i,j}$	η_{old}	$B_{y,savings,i,j}$
1	3.119	34.78%	0.2	2.31
2	3.119	34.43%	0.2	2.25
3	3.119	34.09%	0.2	2.20
4	3.119	33.75%	0.2	2.14
5	3.119	33.41%	0.2	2.09
6	3.119	33.08%	0.2	2.04

7	3.119	32.74%	0.2	1.99
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Example of calculation for Obaapa:

If $y=1$,

$$B_{y,savings,i,j} = 3.051 \times [(0.3478/0.2) - 1]$$

= 2.11 tonnes

Year (y)	$B_{y=1,new,i,survey}$	$\eta_{new,y,i,j}$	η_{old}	$B_{y,savings,i,j}$
1	3.051	33.84%	0.2	2.11
2	3.051	33.50%	0.2	2.06
3	3.051	33.17%	0.2	2.01
4	3.051	32.83%	0.2	1.96
5	3.051	32.51%	0.2	1.91
6	3.051	32.18%	0.2	1.86
7	3.051	31.86%	0.2	1.81

Determination of emission reductions by ICS of year1 during year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO2} + EF_{wf,non\ CO2}) \times N_{y,i,j} \times 0.95$$

Where

$$NCV_{wood\ fuel} = 0.0156 \text{ TJ/tonne}$$

$$f_{NRB,y} = 0.8197$$

$$EF_{wf,CO2} + EF_{wf,non\ CO2} = 112 + 26.23 = 138.23 \text{ tCO}_2/\text{TJ}$$

Example of calculation for Gyapa project stove ($N_{y,i,j} = 18,893$):

If $y=1$,

$$ER_{y,i,j} = 2.31 \times 0.0156 \times 0.8197 \times 138.23 \times 18,893 \times 0.95$$

$$= 73,131 \text{ tCO}_2$$

Year	$B_{y,savings,i,j}$	$NCV_{wood\ fuel}$	$f_{NRB,y}$	$EF_{wf,CO2} + EF_{wf,non\ CO2}$	$N_{y,i,j}$	$ER_{y,i,j}$
1	2.31	0.0156	0.8197	138.23	18,893	73,131
2	2.25	0.0156	0.8197	138.23	18,893	71,410
3	2.20	0.0156	0.8197	138.23	18,893	69,707
4	2.14	0.0156	0.8197	138.23	18,893	68,020

5	2.09	0.0156	0.8197	138.23	18,893	66,350
6	2.04	0.0156	0.8197	138.23	18,893	64,697
7	1.99	0.0156	0.8197	138.23	18,893	63,060

Example of calculation for Obaapa project stove ($N_{y,i,j} = 1209$):

If $y=1$,

$$ER_{y,i,j} = 2.11 \times 0.0156 \times 0.8197 \times 138.23 \times 1209 \times 0.95$$

$$= 4,286 \text{ tCO}_2$$

Year	$B_{y,savings,i,j}$	$NCV_{wood\ fuel}$	$f_{NRB,y}$	$EF_{wf,CO2}$ + $EF_{wf,non\ CO2}$	$N_{y,i,j}$	$ER_{y,i,j}$
1	2.11	0.0156	0.8197	138.23	1,209	4,286
2	2.06	0.0156	0.8197	138.23	1,209	4,181
3	2.01	0.0156	0.8197	138.23	1,209	4,077
4	1.96	0.0156	0.8197	138.23	1,209	3,974
5	1.91	0.0156	0.8197	138.23	1,209	3,873
6	1.86	0.0156	0.8197	138.23	1,209	3,772
7	1.81	0.0156	0.8197	138.23	1,209	3,672

The total ER estimation combined for all Project activity Instances i.e for Gyapa and Obaapa is as below

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
Year 2023-24 (16-January-2023 – 15-January-2024)	77,417	0	0	77,417
Year 2024-25 (16-January-	75,591	0	0	75,591

2024 – 15-January-2025)				
Year 2025-26 (16-January-2025 – 15-January-2026)	73,784	0	0	73,784
Year 2026-27 (16-January-2026 – 15-January-2027)	71,994	0	0	71,994
Year 2027-28 (16-January-2027 – 15-January-2028)	70,223	0	0	70,223
Year 2028-29 (16-January-2028 – 15-January-2029)	68,469	0	0	68,469
Year 2029-30 (16-January-2029 – 15-January-2030)	66,732	0	0	66,732
Total	504,210	0	0	504,210

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass

Source of data	PP has calculated f_{NRB} by determining the share of renewable and non-renewable woody biomass in the total quantity of woody biomass consumption for the country/region
Value applied:	0.8197 (calculated as per meth tool)
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante. The default value as per the Methodological tool: Calculation of the fraction of non-renewable biomass (Tool 30 version 4.0. As per of methodological tool, the value is calculated as per the following Equation (1), $f_{NRB,y} = NRB/(NRB+RB)$
Purpose of Data	Calculation of emission reductions
Comments	If the future project activity instances are expanded in other parts of Africa (other than Ghana), then this parameter value will be calculated during future project activity instances inclusion.

Data / Parameter	NCV _{wood fuel}
Data unit	TJ/tonne
Description	The net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction
Value applied:	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	EF _{wf,CO2}
------------------	----------------------

Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in the baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	EF _{wf,non CO2}
Data unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for the use of wood fuel in the baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied:	26.23
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	η _{p1}
Data unit	Percentage

Description	The efficiency of the Gyapa project stove at the start of the project activity.
Source of data	Manufacturer's specification
Value applied	Efficiency of Gyapa project stove = 37.00 %
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	This efficiency is for the cook stove model considered for project activity instances implemented to date. If the manufacturer or cook stove model changes, the respective model efficiency will be considered for ER calculations

Data / Parameter	η_{p2}
Data unit	Percentage
Description	The efficiency of the project stove at the start of the project activity.
Source of data	Manufacturer's specification
Value applied	Efficiency of project Obaapa Clean stove = 36.00 %
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	This efficiency is for the cook stove model considered for project activity instances implemented to date. If the manufacturer or cook stove model changes, the respective model efficiency will be considered for ER calculations

Data / Parameter	DF_n
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Data unit	Fraction
Description	Discount factor to account for efficiency loss of project cookstove per year of operation (fraction).
Source of data	As per methodology, the default value of 0.99 efficiency loss per year
Value applied	0.99
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of $\eta_{\text{new},y,i,j}$
Comments	-

Data / Parameter	Leakage factor
Data unit	Fraction
Description	Discount factor to account for leakage
Source of data	Methodological default value
Value applied	0.95
Justification of choice of data or description of measurement methods and procedures applied	In-line with the applied methodology VMR0006, ver. 1.1 Section 8.3.
Purpose of Data	Calculation of emission reductions
Comments	-

6.2 Data and Parameters Monitored

Data / Parameter	$N_{y,j,j}$
Data unit	Number

Description	Number of project devices of type I and batch j operating during year y						
Source of data	Monitoring						
Description of measurement methods and procedures applied	Measured directly or based on a representative sample. As per methodology, the minimum sample size will be determined in which case compliance with 90/10 confidence precision is not obligatory.						
Frequency of monitoring/recording	At least once every two years						
Value applied	<table border="1"> <tr> <td>Gyapa</td><td>18,893</td></tr> <tr> <td>Obaapa</td><td>1,209</td></tr> <tr> <td>Total</td><td>20,102</td></tr> </table>	Gyapa	18,893	Obaapa	1,209	Total	20,102
Gyapa	18,893						
Obaapa	1,209						
Total	20,102						
Monitoring equipment	Sample survey						
QA/QC procedures applied	-						
Purpose of data	Calculation of emission reductions						
Calculation method	The proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value						
Comments	PP has distributed one ICS per household and the same is followed for each project activity instance.						

Data / Parameter	$\eta_{\text{new},y,i,j}$
Data unit	Fraction
Description	The efficiency of the improved cook stove type i and batch j (i.e Gyapa project stove and Obaapa Clean stove respectively) determined as per equation 5 of methodology during year y
Source of data	Calculation
Description of measurement methods and procedures applied	Select Option V from the methodology i.e “Efficiency of the improved cookstoves can also be estimated ex-ante using equation 5 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”. However, the grouped project activity provides the option to use a different model of cookstoves for new future project activity instances to be included, hence this parameter value may vary based on a different model of future instances.

Frequency of monitoring/recording	Once at the time of project stove installation																																
Value applied	For ex-ante calculation, the value of - Gyapa project stove below is applied <table border="1"> <thead> <tr> <th>Year(y)</th><th>$\eta_{new,y,i,j}$</th></tr> </thead> <tbody> <tr><td>1</td><td>34.78%</td></tr> <tr><td>2</td><td>34.43%</td></tr> <tr><td>3</td><td>34.09%</td></tr> <tr><td>4</td><td>33.75%</td></tr> <tr><td>5</td><td>33.41%</td></tr> <tr><td>6</td><td>33.08%</td></tr> <tr><td>7</td><td>32.74%</td></tr> </tbody> </table> - Obaapa Clean stove below is applied <table border="1"> <thead> <tr> <th>Year(y)</th><th>$\eta_{new,y,i,j}$</th></tr> </thead> <tbody> <tr><td>1</td><td>33.84%</td></tr> <tr><td>2</td><td>33.50%</td></tr> <tr><td>3</td><td>33.17%</td></tr> <tr><td>4</td><td>32.83%</td></tr> <tr><td>5</td><td>32.51%</td></tr> <tr><td>6</td><td>32.18%</td></tr> <tr><td>7</td><td>31.86%</td></tr> </tbody> </table>	Year(y)	$\eta_{new,y,i,j}$	1	34.78%	2	34.43%	3	34.09%	4	33.75%	5	33.41%	6	33.08%	7	32.74%	Year(y)	$\eta_{new,y,i,j}$	1	33.84%	2	33.50%	3	33.17%	4	32.83%	5	32.51%	6	32.18%	7	31.86%
Year(y)	$\eta_{new,y,i,j}$																																
1	34.78%																																
2	34.43%																																
3	34.09%																																
4	33.75%																																
5	33.41%																																
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3	33.17%																																
4	32.83%																																
5	32.51%																																
6	32.18%																																
7	31.86%																																
Monitoring equipment	Not Applicable																																
QA/QC procedures applied	Not Applicable																																
Purpose of data	Calculation of emission reductions																																
Calculation method	The calculation is to be performed using the equation below: $\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$																																
Comments	A default value of 0.99 efficiency loss per year can be considered																																

	In case of ICS model changes for new project activity instances, the efficiency of the new ICS model should be determined during the verification and the same will be considered for that project activity instance.
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Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes
Description	Annual quantity ²⁴ of woody biomass used by improved cook stoves in tonnes per device of type i and batch j, determined in the first year of the implementation of the project through a sample survey
Source of data	Monitoring survey
Description of measurement methods and procedures applied	The sample size was selected in line with the guidelines provided in methodology Section 8.4 option (b). Determined in the first year of the introduction of the devices (e.g., during the first year of the crediting period, $y=1$) through measurement campaigns at representative households and/or sample surveys. The conditions necessary for the use of sample surveys based solely on questionnaires or interviews (without measurement campaigns) have been found to be satisfied i.e. (a) Baseline cookstoves have been completely decommissioned, and only improved cookstoves are exclusively used in the project households. (b) In the monitoring survey, no multiple devices were found to be in use in the project households. Therefore, there is no need to differentiate the quantity of firewood used by each device, as only a single project device that consumes firewood is in use.
Frequency of monitoring/recording	Determined in the first year of project implementation
Value applied	1. For Gyapa project stove – 8.55 kg/device/day or equal to 3.119 tonnes/device/year. (1.424 Kg charcoal / device / Day) 2. For Obaapa Clean stove – 8.36 kg/device/day or equal to 3.051 tonnes/device/year. (1.393 Kg charcoal / device / Day)

²⁴ A default wood to charcoal conversion factor of 6 kg of firewood per kg of charcoal has been used in line with paragraph 35 of AMS II.G, version 11

Monitoring equipment	Weighing Scale
QA/QC procedures applied	Calibration of weighing scales used for measuring the charcoal was done in-house before start using on site
Purpose of data	Calculation of emission reductions
Calculation method	This is monitored value
Comments	No comments

Data / Parameter	$\eta_{old,i,j}$
Data unit	Fraction
Description	The efficiency of baseline cookstove
Source of data	Methodological default value
Description of measurement methods and procedures applied	During the baseline survey, it is observed that the baseline device was a three-stone fire using charcoal, or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney, hence default value of 0.2 is considered for Efficiency of the baseline cookstove in line with the applied methodology.
Frequency of monitoring/recording	Fixed for each individual household at the time of project implementation.
Value applied	0.2
Monitoring equipment	NA
QA/QC procedures applied	NA
Purpose of data	Calculation of emission reductions
Calculation method	NA
Comments	

Data / Parameter	Life Span
Data unit	Year

Description	Operating lifetime of project device for projects opting Equation 5 for determining project stove efficiency
Source of data	Manufacturer's specification
Description of measurement methods and procedures applied	Measured directly or based on a representative sample. As per methodology, the minimum sample size will be determined in which case compliance with 90/10 confidence precision is not obligatory.
Frequency of monitoring/recording	Once at the time of the new model of Project stove installation
Value applied	7 (Applicable for Gyapa & Obappa models)
Monitoring equipment	NA
QA/QC procedures applied	NA
Purpose of data	Calculation of $\eta_{new,y,i,j}$
Calculation method	NA
Comments	

6.3 Monitoring Plan

For measurement/monitoring of two parameters based on a representative sample. The simplified approach proposed in option (b) under Section 8.4 of the methodology have been used for determining the minimum sample size in which case compliance with 90/10 confidence precision is not obligatory.

Monitored Parameters

The project involves the Introduction of an improved cook stove as energy efficiency measures:

No.	Monitoring Parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,j,k}$ Number of project devices of type i and batch j operating during year y	The proportion of ICS is still in operation.	Proportion	Biennially (at least once in two years)

2	$B_{y=1,new,i,j,survey}$ Quantity of charcoaly biomass used by project devices in tonns per device of type i. and batch j	Daily consumption of charcoal biomass per ICS	Mean value	in the first year of project implementation
3	Life Span	Manufacturer's specification	Number of Years	Once at the time of the new model of Project stove installation

Target Population

The target population is the complete set of appliances (ICS) deployed under the project.

Sampling Method

The project involves the distribution of ICS throughout the project area thereby replacing traditional cooking devices. The population is homogeneous in nature i.e., common technology with similar operating characteristics and dispersed in the country. Since the characteristics of the population (for example quantity of biomass consumed per household) are more similar. Therefore, a simple random sampling technique is being used to calculate the samples. Samples have been drawn by using the random number generator.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,j,j}$ Visual inspection of the premises to see if ICS is operational and in use.
Interview with the end user if required to verify that ICS is still in use (Yes/No)

$B_{y=1,new,i,j,survey}$ Interview with the end user and estimate the daily consumption of charcoal of ICS (Weekly/Daily consumption of charcoal)

Sample Size

A simplified approach is being used for determining the minimum sample size for parameters listed under Section 9.2 of the methodology in which case it is not requisite for the sample size to meet confidence/precision requirements.

The project proponent has chosen to use the same sample to monitor both parameters. According to the Standard for sampling and surveys for VCS project activities and programme of activities, if there is more than one parameter to be estimated, then a sample size calculation should be done for each of them. Then either the largest number for the sample size is chosen as a sampling effort with one common survey, or separate sampling efforts and surveys are undertaken for each parameter. Therefore the project proponent have sample a combination of two parameters i.e. $N_{y,j,j}$ and $B_{y=1,new,i,j,survey}$ in the same sample. Sampling more than one

parameter within the same sample (household) helps reduce travel needs for monitoring and the associated costs.

Oversampling is strongly encouraged, not only to compensate for any attrition, outliers, or non-response associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved, and additional sampling efforts would be required. The sample size shown below in this section have been adjusted upwards to account for non-responses, Project proponent shall determine the appropriate non-responses rate based on previous experience.

Sampling Frame

The sampling frames shall be defined below. Overall, ICS have the same group of end users which is household users, thus it is expected that the geographical locations do not influence the parameter of interest. Therefore, all monitoring parameters can be assumed to be highly homogeneous for each ICS model regardless of how the end user group and distribution/installation location are defined.

Data to be collected

Field Measurement:

The table below summarizes field measurement data requirements:

Parameter	Timing (indicative)	Frequency as required by the methodology	Methods to be applied
$N_{y,j,j}$	Monitoring occurs every 12 to 24 months	Biennially	Visits to the premises, visual inspection, and interview with ICS end-user.

$N_{y,j,j}$

The survey involves a visual inspection of the premises to see if ICS is operational and in use. An interview with the end user also been conducted if required to verify that ICS is still in use. Based on the response received from the end user, the value of the parameter has been calculated.

$B_{y=1,new,l,j,survey}$	Monitoring occurred within the first year after installation	The first year of installation	Interview with the end user for determining the average quantity of charcoal used in the project stove per week/day. Measurement campaigns for estimation of consumption of charcoal in project households. (Charcoal fuel quantity)
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$B_{y=1,new,l,j,survey}$

This survey involves interviews with the end user and estimating the daily consumption of charcoal of ICS (Weekly/Daily consumption of charcoal). The staff visited the site and ask each household about their daily consumption of charcoal for the project stove. A quantity of charcoal made from the Charcoal stock available with the user and then weighed using a weighing scale. The weight of charcoal used in survey households then noted down similarly and then averaged for the final value of the parameter.

Soft copies of the surveys have been kept in a database with a backup. The original survey questionnaires filled by the survey team and signed by the users are stored in the project office. A backup of the survey results also be stored on an electric medium. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of credits for the project activity, whichever is later.

Quality Assurance/Quality Control

The project proponent considers the non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to each parameter depends on the cost of each data collection approach and the logistics required. The project proponent determines the most effective measure for each parameter from the following list:

- Oversampling: Randomly draw a sample more than the calculated number (say 20%) and collect data for each parameter
- Buffer Group: Randomly draw additional samples (say 20%) and collect data from only minimum numbers of ICS as per sample size calculation. If this would not result in the required sample size data would be collected from the additional ICS that were selected in the sample.

The sampling plan has the following procedures in place to ensure good-quality data. The project proponent ensures that field personnel have reviewed, understand, and agreed to follow the monitoring plan procedures, including provisions for maximizing response rates and documenting out-of-population cases, refusals, and other sources of non-response. A quality control and assurance strategy has been documented. Quality control and assurance strategies include addressing non-sampling errors, such as non-response or bias from the interviewer. The project proponent or a competent third party designated by the project proponent with the proper skills have trained the monitoring personnel on how to properly survey households to prevent bias from the interviewer. In the case a household refuses to participate, another household will be chosen at random. To reduce interviewer bias, a good questionnaire design and well-tested questionnaires are being used. The surveyor at the site gets training from EKI for the monitoring survey, which includes interviewing the beneficiaries, weighing the charcoal with weighing equipment. Additionally, gives weighing machine calibration instructions.

The sample data for the mean value parameter is continuous and therefore the presence of outliers is possible. To identify and address outliers for the parameter, outliers are defined as those data points with values greater than three standard deviations from the mean of the sample. Data points identified as outliers are examined further to correct for possible transcription and data entry errors but are omitted from the analysis if no such administrative errors exist.

Data collection and management methods

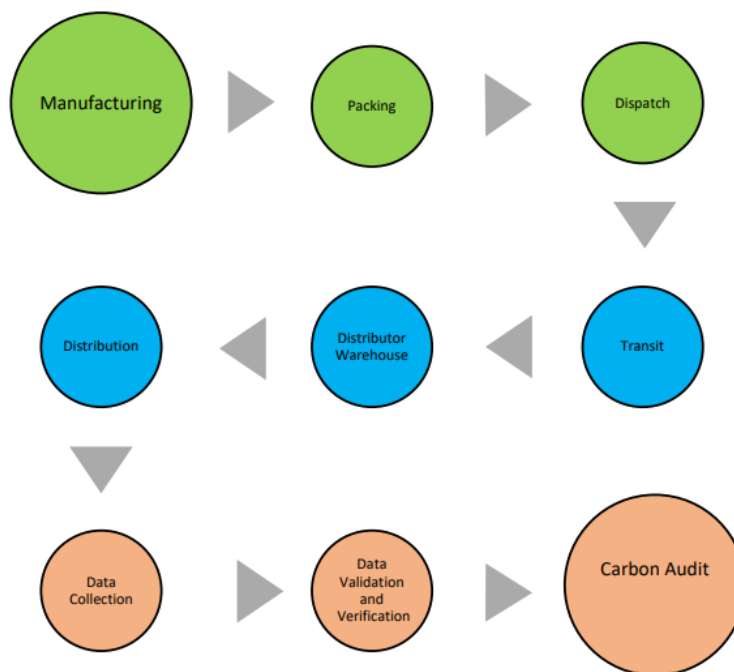
The methods for Distribution, Monitoring, and reporting data and parameters are described below.

Each distributed ICS is equipped with a label stating the unique serial number which is used for references and monitoring. The unique serial numbers provide the basis for monitoring activities and prevent instances of double counting. Other than the unique serial number, the following is the minimum information that will be captured for each project device in order to be eligible for inclusion in this project:

1. Date of distribution
2. Geographic area of distribution
3. Model of project technology distributed
4. Quantity of project technologies distributed
5. Name and address of the recipient/end-user

The ICS is transported by the distributing entity to the Project site. Users receive the ICS through distributing entities only. The ICS data collection and management system uses a safe data storage system using the mobile app. ICS distribution data are also being collected using an offline mechanism i.e., manual entry. Developing an app allowed much larger-scale data collection & management. Additionally, the app offers details on the activities of distribution partners and accumulated data sets.

The ICS flow is described in the following figure:



Monitoring questionnaires

The distributing entities record each sample ICS user's contact information and pertinent data on standardized monitoring questionnaires concerning the corresponding serial number of the ICS that was distributed.

The interviewer can fill out the monitoring questionnaire manually or using an app-based electronic device (e.g., mobile phone) at each sample household. Data recorded through a manual process will be recorded and transferred into the electronic database.

Stoves Users are requested to provide the required information to the interviewer. Users can share their background information in several ways, including by speaking directly with designated focal points who may reside in their communities. This includes:

- Stove Serial No. /Unique ID, Date of ICS Receipt/onset of usage, Address, Beneficiary Name, etc.
- Improved Cookstove Specific Information such as the functioning of ICS, benefits observed from the use of improved cook stoves
- Traditional Cookstove Specific uses information
- Other Cooking fuel-specific information – LPG, Kerosene, etc.
- Fuel use pattern of your household – Purpose of fuel use
- Fuel Details – Quantity, sources, Cost, Quantity, Problem in collecting fuel, etc.

Calibration procedures for monitoring equipment:

To measure the weight of charcoal consumed by the project stove in a day/week, weighing scales are being used. The weighing scales used for monitoring purposes are being calibrated before starting the survey, as per the national/international standard, or as specified in the user manual of the respective weighing scale.

Analysis

The project proponent will manage a project database that includes the following data that can be directly attributable within the project, thereby allowing unambiguous determination of the emission reductions attributable to each project:

- A list of households participating in the project, including name, community/location, distribution/installation date, and unique serial number;
- Where replacements are made, assurance that the efficiency of the new ICS is similar to the specified.

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values are then being factored into the emissions reduction calculations and result in the request for the issuance of VCUs. The parameters are applied for emission reduction calculations. The stoves that are not in use have been excluded from emissions reduction calculations and are not being counted towards the total number of ICS in operation during the monitoring period.

Implementation

The project participant followed the simple random sampling approach and the minimum sample size have been determined as per the following guidelines:

- Project target population < 300: Minimum sample size 30
- Project target population 300 – 1000: Minimum sample size 10% of group size
- Project target population > 1000: Minimum sample size 100

The simplified approach will be used for determining the minimum sample size for parameters to be monitored in which case it is not requisite for the sample compliance with 90/10 confidence precision is not obligatory.

As per the requirement of applied VCS methodology VMR0006, a minimum sample size of 100 is required to be surveyed for the project population greater than 1000. However, to get more accurate and reliable results, EKI has selected a bigger number of samples to conduct the survey.

Sampling for emission reduction calculation and elaboration of the monitoring report occurs at the end of each monitoring period. This sampling has been conducted by trained personnel from the project proponent or an experienced third-party entity. The credentials and/or training

materials for the sampling personnel will be provided to the VVB at verification. The maximum length of one monitoring period will be two years (duration, not calendar years), with an option for annual or biennial monitoring. The project proponent is responsible for managing household data collection and entry into the project database. Field personnel have received training on how to properly deal with surveying techniques and reduce errors and checked that there is no conflict of interest among those involved in data collection and analysis. If there is a conflict of interest, the personnel will not be allowed to participate in data collection and analysis. The project proponent will record the start and end dates of each monitoring period and calculate the emission reductions attributable to each monitoring period.

Survey results:

The following parameters were obtained through sampling:

1. $N_{y,i,j}$: Number of stoves in operation
2. $B_{y=1,new,i,j,survey}$: Quantity of charcoal used by improved cookstoves

125 samples were selected (considering 20% non-response) from each cookstove model (i.e., Gyapa project stove and Obaapa Clean stove) for all monitoring parameters by section 8.4 option (b) of the applied methodology VMR0006. The survey team has surveyed 100 samples each on the ground from the 125 selected samples. The objective was to obtain reliable and unbiased estimates of the monitoring parameters.

A single homogeneous population was considered for both parameters.

The following table summarizes the sample sizes and results.

Monitored Parameter	Gyapa	Obaapa
Sample size	100	100
Average Quantity of charcoal used by improved cookstoves ($B_{y=1,new,i,j,survey}$) Kg/day/stove	1.424	1.393

Appropriate record-keeping procedures have been implemented to ensure that each monitoring period data set can be transparently attributed to its corresponding project, preventing any occurrences of double counting. An internal review of the project database is being conducted to determine the current status of each device — the duration of previous monitoring periods, the households delivering monitoring data, and current verification activities.

(i) Assessment for Leakage

The methodology provides a net-to-gross adjustment factor of 0.95 to account for leakages, hence the surveys are not required to determine leakage.

The other source of leakage occurs if equipment currently being utilized is transferred from outside the boundary to the project activity. All ICSs in the project are newly manufactured/assembled or newly installed, thus no leakage is applicable.

(ii) Monitoring Reporting

The project proponent assesses all monitoring data and produce a monitoring record for the VVB to verify the corresponding to the preceding monitoring period. This record presents the data relating to the survey results conducted for the respective monitoring period.

The purpose of the monitoring plan is to ensure successful monitoring of the emission reductions of the proposed project during its crediting period. The overall monitoring is being managed by the project implementer EKI Energy Services Limited (EKIESL)

Activities and performance related to emissions reduction are monitored by EKIESL. The PP proposed the following roles and responsibilities for data monitoring, collection, data archiving, and calibration of equipment for these project activity instances.

The monitoring activities involves data collection during distribution as well as usage information post-distribution. The data collected during distribution also involves information about the stove, the end user, and the location to enable one to uniquely identify each ICS unit and avoid double counting.

Repair and Maintenance - EKIESL/its representative planning to establish a service centre at the project location where end users can repair the cookstoves and can use them further. The repair and maintenance will be followed as per the manufacturer's recommendation. This will ensure that the cookstove will operate for a longer period. After 7 years of lifetime, a report from the manufacturer/maintenance agency will be provided to confirm the extended life of the cookstove. The project will claim credits beyond 7 years only for a balanced year till the cookstove is replaced by new cookstoves free of cost only and will be updated in database accordingly.

In case, the end user experiences any issues with the ICS, he or she can call to the customer service number available on the box ICS, to report the issues. After receiving a complaint, the maintenance team will take the necessary action immediately to resolve the issue.

EKIESL/its representatives are distributing the stoves, as well as carrying out the monitoring activities that occur during the distribution of stoves. EKIESL or their associates ensure that the distributors are trained on how to capture the end-user data. The project activity implementer is fully responsible to ensure the correct distribution process and data gathering is followed.

EKIESL team also visits the site regularly to ensure proper use of stoves, take feedback and make the end user aware of their health and environmental sustainability as well as the benefits of saving time and money with the use of ICS. The team also checking the grievances of the local stakeholders from the project activity and trying to resolve those issues on the ground. As a Part of Internal Audit team visits random households and perform basic audit to ensure its usage, and its physical condition. All the details of internal audits were feed in internal Audit register, if in case beneficiary needs any help regarding ICS or having problem in ICS then ICS is repaired or replaced free of cost.

The following information are being recorded at the time of distribution of ICS to the user:

- Name of the User

- Total Family Members
- Location/Address
- Installation Date
- Unique ID

Once ICS is distributed, monitoring activities involves selecting a sample of the stove from the distribution record and visiting the premises where these stoves are located to monitor key parameters pertinent to ER Calculation. The individuals carrying out the monitoring activities follows instructions provided during training, to check and record the following key parameters:

- Check if project stoves are operational and in use
- Check if there is any ongoing use of replaced stoves.

Check biomass consumption of new cookstove Data collecting & handling are conducted in a transparent way to secure the high quality of recording and storing of data. Data collected and monitored are stored electronically in a secure and retrievable manner for at least two years after the end of the project crediting period. Uncertainty related to data handling (if any) would be rectified, if necessary, by revising monitoring procedures. The changes would be approved by a VVB.

The roles and responsibilities of different participators are proposed to be as follows:

Entity/Person	Roles & Responsibilities
EKIESL	✓ VCS documentation development ✓ Communicate with VERRA for all VCS related matters ✓ Communicate with VVB for validation & verification ✓ Identify local partners for local representation and support ✓ Identity suitable ICS manufacturers ✓ Design the distribution plan for on-ground implementation of ICS ✓ Design the monitoring system (including data collection in an electronic database) ✓ Training and development of local resources (as may be required) ✓ Regular Monitoring and quality assurance of data ✓ Monitor & supervise on-ground distribution, etc. ✓ Annual Review meeting with Local Teams and other associated entities to ensure continuous improvements in the project activity
Distributor / Operational Team (Local Representative)	✓ To support & assist EKIESL in achieving the stated goal of the project activity ✓ Assist and facilitate EKIESL in baseline data collection, efficiency testing, data of biomass consumption for new ICS

	through sampling survey, etc. ✓ To be the local representative and communicate, act and engage in implementation activities of project activity ✓ To identify and deploy suitable local teams and resources for ICS distribution, data collection, regular spot-checks at households, etc. ✓ Assist and facilitate EKIESL during the training of the Operational Team and capacity building across the end users. ✓ To bridge any gap between end users of ICS and EKIESL to ensure the smooth operation of the program to achieve the overall objective of the project activity ICS Distributor(s)–who are responsible for deploying the ICS at the household and registering the end user in the digital tool. They are also responsible for giving demonstrations of ICS use, discussing with the end users on its benefits, and educating them on the continued use of ICS and discontinuation of the use of conventional/traditional open firing practice. They also conducts a periodic check at households to gather feedback, address any technical concerns, collect necessary data and transmit the information.
ICS Manufacturer or Supplier	✓ To design and develop ICS as per the required standard ✓ To provide technical parameters and manufacturer's efficiency etc. at the time of supply ✓ To educate and train the EKIESL team on the installation (delivery), uses, handling, etc. of the ICS in households.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$		
Data unit	Number		
Description	Number of project devices of type I and batch j operating during year y		
Value applied:	Number of ICS distributed	The proportion of ICS operational (based on sample survey)	Number of ICS operational during the monitoring period

	20,102 (Till the end of this monitoring period i.e., 30-November-2022)	100%	20,102
Comments	The proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value. The survey will be conducted at least once every two years basis after the implementation of the project activity instance.		

Data / Parameter	$\eta_{(new,y,i,j)}$								
Data unit	Percentage								
Description	The efficiency of the improved cookstove type i and batch j determined using equation 5 of applied methodology during year y								
Value applied:	- Gyapa project stove below is applied <table border="1"> <tr> <th>Year(y)</th><th>$\eta_{new,y,i,j}$</th></tr> <tr> <td>1</td><td>34.78%</td></tr> </table> - Obaapa Clean stove below is applied <table border="1"> <tr> <th>Year(y)</th><th>$\eta_{new,y,i,j}$</th></tr> <tr> <td>1</td><td>33.84%</td></tr> </table>	Year(y)	$\eta_{new,y,i,j}$	1	34.78%	Year(y)	$\eta_{new,y,i,j}$	1	33.84%
Year(y)	$\eta_{new,y,i,j}$								
1	34.78%								
Year(y)	$\eta_{new,y,i,j}$								
1	33.84%								
Comments	No comments								

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes
Description	Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j, determined in the first year of the implementation of the project through a sample survey

Value applied:	1. For Gyapa project stove – 8.55 kg/device/day or equal to 3.119 tonnes/device/year. 2. For Obaapa Clean stove – 8.36 kg/device/day or equal to 3.051 tonnes/device/year. Estimated based on the survey result
Comments	The survey was conducted to determine this value in the first year of project implementation.

7.2 Baseline Emissions

The methodology does not account for baseline emissions separately but instead quantifies net emission reductions achieved by the project. Please refer to Section 7.5 below.

7.3 Project Emissions

The methodology does not account for project emissions separately but instead quantifies net emission reductions achieved by the project. Please refer to Section 7.5 below.

7.4 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology.

7.5 Net GHG Emission Reductions and Removals

PP has calculated the emission reduction for each ICS installed under the project activity separately. In this section example of ER calculation of one ICS (stove serial number EKIGH-22/L1/0001446) is provided:

Date of installation: 16-January-2023

Location: Ghana

Cookstove Model: Gyapa

For the year 2023, vintage for the stove: Vintage 1

No. of days for vintage 1: 44 (16-January-2023 to 28-February-2023) *(inclusive of both the dates)*

Year fraction: $44/365 = 0.121$

Determination of efficiency of ICS during year y

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where

$$\eta_p = 37\%$$

$$DF_n = 0.99$$

Example of calculation for Gyapa:

$$\text{If } y = 1$$

$$\eta_{new,y,i,j} = 37\% \times (0.99)^{(1-1)} \times 0.94$$

$$= 34.78\%$$

Determination of the quantity of charcoal that is saved in tonnes per ICS during year y

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right)$$

$$\text{If } y = 1,$$

$$B_{y=1,i,j,survey} = 3.119$$

$$B_{y,savings,i,j} = 3.119 \times [(0.3478/0.2) - 1]$$

$$= 2.31 \text{ tonnes}$$

Period	$B_{y=1,new,i,survey}$	$\eta_{new,y,i,j}$	η_{old}	$B_{y,savings,i,j}$
16-January-2023 to 28-February-2022(inclusive of both the dates)	3.119	34.78%	0.2	2.31

Determination of emission reductions

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where,

$$NCV_{wood\ fuel} = 0.0156 \text{ TJ/tonne}$$

$$f_{NRB,y} = 0.8197$$

$$EF_{wf,CO_2} + EF_{wf,non\ CO_2} = 112 + 26.23 = 138.23 \text{ tCO}_2/\text{TJ}$$

If $y = 1$

$$ER_{y,i,j} = 2.31 \times 0.0156 \times 0.8197 \times 138.23 \times 1 \times 0.95 \times 0.121$$

$$= 0.467 \text{ tCO}_2$$

The total emission reductions for all the project activity instances are as below

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
Year 2023 (16-January-2023-- 28-Februrary-2023) (inclusive of both the dates)	4,291	0	0	4,291
Total	4,291			4,291

<u>Ex-ante emissions reductions /removals</u>	<u>Achieved emissions reductions /removals</u>	<u>Percent difference</u>	<u>Justification for the difference</u>
8,517	4,291	-49.62%	All the project ICS were not installed on the first date of monitoring

APPENDIX 1: < SUPPORTING EVIDENCE >

Employment Declaration (Manufacturer)



SUSTAINABLE ENERGY TECHNOLOGIES LIMITED

Engineering, Manufacturing and Distribution of Renewable Energy and Agricultural Technologies (Household & Commercial Clean cookstoves, Solar Lighting and Installation, Water Filters, Biogas Systems & Agricultural Machinery)

University Box 658, University of Ghana, Legon, Accra

TELEPHONES: 0244108268 / 0248737439. Email: info@setechghana.com

Date: 29-03-2023

This is to declare that Sustainable Energy Technologies (SETECH) has employed 33 people for improved cookstove manufacturing to execute Enking orders of improved cook stoves from Sep 2022 onwards. Details of same is as below:

Blue collar Workers :- 25 (20 male and 5 female), Average salary GHS 2000 to 5000 per month.
Staff :- 8 (5 Male and 3 females), Average salary GHS 5000 to 9000 per month. Yours Faithfully,



Lovans Owusu-Takyi

Executive Director

For Sustainable Energy Technologies (SETECH)

Employment Declaration (Distributor)



SUSTAINABLE ENERGY TECHNOLOGIES LIMITED

Engineering, Manufacturing and Distribution of Renewable Energy and Agricultural Technologies (Household & Commercial Clean cookstoves, Solar Lighting and installation, Water Filters, Biogas Systems & Agricultural Machinery)

University Box 658, University of Ghana, Legon, Accra

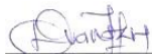
TELEPHONES: 0244108268 / 0248737439. Email: info@setechghana.com

Date: 27-03-2023

This is to declare that Sustainable Energy Technologies (SETECH) has employed 60 part people for improved cookstove Deployment to execute Enking orders of improved cook stoves from Oct 2022 onwards. Details of same is as below:

Blue collar Workers :- 55 Nos (30 male and 25 female), Average salary GHS3500 to GHS5500 per month.
Staff :- 5 Nos (3 Male and 2 female), Average salary GHS7000 to GHS12,000 per month.

Yours Faithfully,



Lovans Owusu-Takyi

Executive Director

For Sustainable Energy Technologies (SETECH)

Training Declaration



SUSTAINABLE ENERGY TECHNOLOGIES LIMITED

Engineering, Manufacturing and Distribution of Renewable Energy and Agricultural Technologies (Household & Commercial Clean cookstoves, Solar Lighting and Installation, Water Filters, Biogas Systems & Agricultural Machinery)

University Box 658, University of Ghana, Legon, Accra

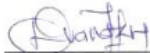
TELEPHONES: 0244108268 / 0248737439. Email: info@setechghana.com

27-03-2023

DECLARATION

This is to declare that Stove distribution team of Sustainable Energy Technologies (SETECH) have trained the users for proper use of the cookstoves during the distribution of project stoves for VCS 3801 Projects . Distribution team also educated/ trained the users towards the benefits of project stove (Improved cookstove), i.e the reduction in the smoke, respiratory problems, improvement in the quality of cooked food etc, It was informed that the project stove consumes less fuel wood, resulting the time saving of the user in collecting the wood as well as saving the environment by reduction in cutting the trees.

Yours Faithfully,



Lovans Owusu-Takyi

Executive Director

For Sustainable Energy Technologies (SETECH)